WORKED SOLUTION

STEP III 2011, Question 1

Pure • Reduction of Order

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q1

The key idea

Part (i) is a separable equation, and it is there for a reason: it is exactly the equation that appears when we substitute $y = ze^{-x}$ into $(*)$. The term in z itself cancels, leaving an equation involving only z' and z'', which is first order in $u = z'$. This is the method of *reduction of order*: once we know one solution (e^{-x}) we can find the other. Part (iii) then runs the same substitution with a right-hand side, which turns into a linear first-order equation solved by an integrating factor.

Solution

Throughout we work on an interval where $x \neq -1$, so we may divide by $x+1$.

Part (i)

Since

$$\frac{x+2}{x+1} = 1 + \frac{1}{x+1},$$

separating the variables (for $u \neq 0$) gives

$$\int \frac{\mathrm{d}u}{u} = \int \left(1 + \frac{1}{x+1}\right) \mathrm{d}x \quad \Longrightarrow \quad \ln|u| = x + \ln|x+1| + c.$$

Exponentiating, $|u| = e^c\,|x+1|\,e^x$. The sign and the factor e^c combine into a single arbitrary constant, and $u = 0$ is also a solution, so the general solution is

$$u = C(x+1)e^x, \qquad C \text{ an arbitrary constant.}$$

Part (ii)

With $y = ze^{-x}$ the product rule gives

$$y' = (z' - z)e^{-x},$$
$$y'' = (z'' - z')e^{-x} - (z' - z)e^{-x} = (z'' - 2z' + z)e^{-x}.$$

Substituting into $(*)$ and cancelling the non-zero factor e^{-x}:

$$\begin{aligned} 0 &= (x+1)(z''-2z'+z)+x(z'-z)-z \\ &= (x+1)z''+(-2(x+1)+x)z'+((x+1)-x-1)z \\ &= (x+1)z''-(x+2)z'. \end{aligned}$$

The z terms have cancelled completely. Writing $u = \dfrac{\mathrm{d}z}{\mathrm{d}x}$ and dividing by $x+1$,

$$\frac{\mathrm{d}u}{\mathrm{d}x} - \left(\frac{x+2}{x+1}\right)u = 0,$$

which is a first-order equation for $\dfrac{\mathrm{d}z}{\mathrm{d}x}$, as required. It is precisely the equation of part (i).

Ask yourself Why did the z term vanish? Could we have predicted it?

The coefficient of z is what we get by putting $z=1$, that is, by testing $y=\mathrm{e}^{-x}$ in $(*)$. Since $(x+1)\mathrm{e}^{-x} - x\mathrm{e}^{-x} - \mathrm{e}^{-x} = 0$, $y=\mathrm{e}^{-x}$ is already a solution, so constant z must work and no bare z term can survive. That is why this substitution was chosen.

By part (i), $z' = C(x+1)\mathrm{e}^{x}$. Integrating by parts,

$$\int (x+1)\mathrm{e}^{x}\,\mathrm{d}x = (x+1)\mathrm{e}^{x} - \int \mathrm{e}^{x}\,\mathrm{d}x = x\mathrm{e}^{x} + \text{const},$$

so

$$z = Cx\mathrm{e}^{x} + B.$$

Hence $y = z\mathrm{e}^{-x} = Cx + B\mathrm{e}^{-x}$, and renaming C as A,

$$y = Ax + B\mathrm{e}^{-x},$$

with A and B arbitrary constants. Every solution of $(*)$ arises this way, since any y can be written as $z\mathrm{e}^{-x}$ with $z = y\mathrm{e}^{x}$; so this is the general solution, as required.

Part (iii)

The same substitution changes only the right-hand side, which becomes $(x+1)^2$ divided by e^{-x}:

$$(x+1)z'' - (x+2)z' = (x+1)^2\mathrm{e}^{x}.$$

With $u = z'$ and dividing by $x+1$,

$$\frac{\mathrm{d}u}{\mathrm{d}x} - \left(\frac{x+2}{x+1}\right)u = (x+1)\mathrm{e}^{x}.$$

This is linear. From part (i), $(x+1)\mathrm{e}^{x}$ solves the homogeneous equation, so its reciprocal $\dfrac{\mathrm{e}^{-x}}{x+1}$ is an integrating factor. (It is $\exp(-\int \frac{x+2}{x+1}\,\mathrm{d}x)$ up to sign.) Multiplying through,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{u\,\mathrm{e}^{-x}}{x+1}\right) = 1 \quad\Longrightarrow\quad u = (x+1)\mathrm{e}^{x}(x+k)$$

for a constant k. So

$$z' = (x^2 + x)\mathrm{e}^x + k(x+1)\mathrm{e}^x.$$

The second term integrates to $kx\mathrm{e}^x$ as in part (ii). For the first, integrating by parts twice,

$$\begin{aligned}\int (x^2 + x)\mathrm{e}^x \,\mathrm{d}x &= (x^2 + x)\mathrm{e}^x - \int (2x+1)\mathrm{e}^x \,\mathrm{d}x \\ &= (x^2 + x)\mathrm{e}^x - (2x+1)\mathrm{e}^x + 2\mathrm{e}^x + \text{const} = (x^2 - x + 1)\mathrm{e}^x + \text{const}.\end{aligned}$$

Therefore $z = (x^2 - x + 1)\mathrm{e}^x + kx\mathrm{e}^x + B$, and

$$y = z\mathrm{e}^{-x} = x^2 - x + 1 + kx + B\mathrm{e}^{-x}.$$

The $-x$ merges with the arbitrary kx, so writing $A = k - 1$ the general solution is

$$y = x^2 + 1 + Ax + B\mathrm{e}^{-x},$$

with A and B arbitrary constants.

Alternatively, since the right-hand side is a quadratic, we could try a particular integral $y = px^2 + qx + r$. Substituting gives $(x+1)2p + x(2px + q) - (px^2 + qx + r) = px^2 + 2px + 2p - r$, and matching with $x^2 + 2x + 1$ gives $p = 1$, $r = 1$, with q free (because x is a complementary function). Adding the complementary function from part (ii) gives the same answer.

Checks

Substitute $y = x^2 + 1$ directly: $y' = 2x$, $y'' = 2$, so the left-hand side is $2(x+1) + 2x^2 - x^2 - 1 = x^2 + 2x + 1 = (x+1)^2$, as needed. Numerically, with random A, B and random $x \in (-0.9, 3)$, finite-difference derivatives of $y = x^2 + 1 + Ax + B\mathrm{e}^{-x}$ satisfy the equation to within about 3×10^{-6}; the same test confirms parts (i) and (ii).

Common slips

- Getting y'' wrong: it is $(z'' - 2z' + z)\mathrm{e}^{-x}$, with a *plus* on the z; a sign error here leaves a stray z term.
- Forgetting to multiply the right-hand side by e^x in part (iii) when cancelling the factor e^{-x}.
- Losing a constant: the final answer needs two arbitrary constants, one from each integration.

Beyond STEP

Reduction of order is a general technique: if y_1 solves a linear second-order equation, putting $y = y_1 v$ always removes the v term and leaves a first-order equation for v'. It is

also the idea behind *variation of parameters*, which produces particular integrals for any right-hand side. Both are treated clearly in Boyce and DiPrima, *Elementary Differential Equations*.

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